

Niki Liao

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Education

Bachelor of Science, Computer Science. GPA: 3.891

Expected Graduation: May 2025

San Jose State University – San Jose, CA.

- Dean's Scholarship Recipient
- President's Scholar (Three Times)
- Scholar Male Athlete Club Sports Award

Master of Science, Computer Science. GPA: 4.0

Expected Graduation: May 2027

San Diego State University – San Diego, CA

Experience

Advantest Inc - San Diego, CA

May 2026 - Aug 2026

AI Application Engineer Intern

- Worked with Advantest **V93000** and **SmarTest** for semiconductor test development.
- Automated **DPSGen** and **FlowGen** workflows using **AI-driven** solutions.
- Improved **test** generation **efficiency** by reducing manual engineering tasks.

AGAPEA Gran Canaria SL - Malaga, Spain.

Jun 2025 – Aug 2025

Machine Learning Engineer Intern

- Improved an existing hotel pricing algorithm by integrating a **Random Forest** based machine learning model to predict optimal room prices.
- Leveraged **hotel location** and **booking period data** to simulate real market behavior and enhance pricing accuracy.
- Automated **daily margin updates** by analyzing the volume of purchase requests from partner agencies, enabling **dynamic pricing** adjustments.

Inspira Mexico - San Diego, CA.

May 2024 – Aug 2024

Software Engineer Intern

- Developed a web application that utilizes **AI** to analyze hotel pricing trends and insights for hotel management.
- Optimized database performance and efficiency using **MongoDB**, reducing query execution time by 30%.
- Assisted in designing and implementing **data pipelines** for AI-driven hotel price analysis.
- Collaborated with the development team to refine **backend services**, improving system performance and scalability.

San Jose State University - San Jose, CA.

Jan 2023 – May 2023

CS Teaching Assistant

- Led weekly tutorials for over 30 students in Java, **"Intro to Data Structures."**
- Achieved an impressive **95%** student satisfaction rating, demonstrating strong teaching and communication skills.
- Assisted professors in developing and improving course materials for better comprehension and engagement.

Projects

AI Tutor Study: Hybrid RAG Learning Assistant

Sep 2025 – Jan 2026

- Built an AI tutor using a hybrid RAG architecture to answer course questions from uploaded files and images.
- Implemented document ingestion with OCR, chunking, SHA-256 deduplication, OpenAI embeddings, FAISS, BM25, and RRF ranking.
- Added citation enforcement, prompt-injection defenses, quiz generation, flashcards and study schedule planning.

Image Classifier with Machine Learning

Sep 2023 – Dec 2023

- Trained an image classification model using scikit-learn and the CIFAR-10 dataset.
- Preprocessed image data and used a Support Vector Machine (SVM) to classify objects like cats, cars, and airplanes.
- Evaluated model accuracy and visualized predictions with matplotlib.

La Liga Match Outcome Predictor

May 2023 – Aug 2023

- Built a machine learning pipeline to predict La Liga match outcomes using historical data.
- Created leakage-safe features with rolling stats, Elo ratings, and chronological validation.
- Trained XGBoost, LightGBM, and Logistic Regression models with a Streamlit prediction dashboard.

Technical Skills

Programming Languages: Python, Java, JavaScript, Racket, Ruby, Prolog, Scheme, C, C++, Linux.

Web Development: HTML, CSS, React.js, Node.js.

Databases: MongoDB, MySQL, SQLite.